Structural Patterns in News Headlines

A Computational Linguistic Analysis Using NLP

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NLP Final Project

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The Question

Do news headlines follow stable syntactic templates

that can be modeled and evaluated rigorously?

Focus: linguistic form, not topic semantics.

Goal: interpretable structure + style prediction.

Why Headlines?

Headlines are extreme compression:

- Minimal length, maximal information
- Strong editorial constraints
- High public exposure and downstream impact

NLP relevance:

- Concise generation
- Summarization quality control
- Framing and agency analysis

Core Thesis

Headline writing is not free-form text production; it is a constrained system with reproducible structural regularities.

Hypothesis and Falsification

Hypothesis

News headlines follow stable, reusable syntactic templates that are recoverable from parse-level features at rates above trivial baselines.

Falsification criteria:

- Macro F1 collapses toward majority/random behavior
- No dominant recurring templates in corpus-level analysis
- Large instability across random train/dev/test splits
- Rules cannot be linked to parse evidence

Pipeline Overview

End-to-end workflow

1. RSS headline collection
2. spaCy parsing (POS, dependencies, entities, roots)
3. Structural analysis and diagnostics
4. Rule-based structure and style prediction
5. Gold-standard evaluation + split-stability sweep

Parsing robustness

Model preference hierarchy + normalization + fallback parse selection for difficult casing/formatting conditions.

What "Gold" Means and Why We Need It

Gold-standard labels = human-verified ground truth

Why necessary:

- Prevent circular evaluation
- Measure true error, not self-agreement
- Enable valid precision/recall/F1
- Support fair model comparisons

Dataset design:

- 391 manually labeled headlines
- Train/Dev/Test = 59.6% / 19.2% / 21.2%
- Split chosen to balance:
  - rule learning capacity (train),
  - tuning diagnostics (dev),
  - unbiased final estimate (test)

Without gold annotation, high scores can be meaningless.

Why These Metrics in This Study

Accuracy

- Fraction of headlines whose predicted label equals the gold label
- Useful for top-line editorial reliability

Precision$_c$ / Recall$_c$

- Precision$_c$: when we predict class c, how often are we right?
- Recall$_c$: of true class-c headlines, how many do we recover?

$F1_c$

- Harmonic mean of precision and recall for class c
- Punishes one-sided behavior

Macro F1

- Equal average over classes
- Critical under imbalance (‘simple_clause‘ dominates)
- Prevents "high accuracy, poor minority labels"

Structural and Style Label Space

Structure labels

- question_form
- passive_clause
- coordination
- noun_phrase_fragment
- simple_clause
- other

Style dimensions

- lead_frame
- agency_style
- density_band
- rhetorical_mode

$$\text{signature} = \text{structure} \mid \text{lead_frame} \mid \text{rhetorical_mode}$$

Evaluation Metrics

$$\text{Accuracy} = \frac{\#\text{correct}}{\#\text{total}}$$

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c}$$

$$F1_c = \frac{2PR}{P + R}, \quad \text{Macro F1} = \frac{1}{|C|} \sum_{c \in C} F1_c$$

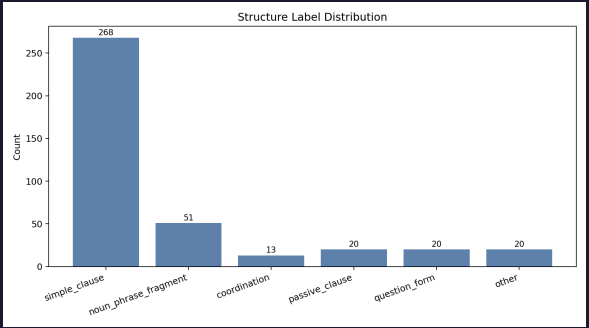
Why Macro F1 matters

- Class imbalance is substantial
- Prevents dominant classes from hiding failures
- Better reflects minority-class behavior

In our context

- Accuracy = top-line headline match rate
- Macro F1 = class-balanced structural reliability

Corpus-Level Findings



Headline profile

- Avg length: 13.7 tokens
- Content-word ratio: 73.2%
- Headlines with verbs: 92.1%
- Active/non-passive: 94.6%
- Named entities present: 95.7%
- Actor/entity-first openings: 81.6%

Finding: headline form is compact, entity-heavy, and strongly action-oriented.

Model Performance Overview

Structure (held-out test)

- Accuracy: 0.759
- Macro F1: 0.594

Baselines (test)

- Majority: 0.639 acc, 0.130 macro F1
- Random: 0.446 acc, 0.187 macro F1

+12.0 pp acc over majority; +31.3 pp over random.

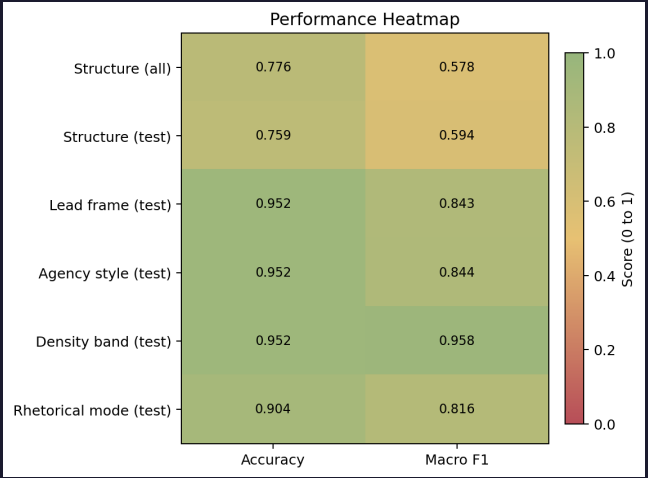
Style Performance (held-out test)

Dimension	Accuracy	Macro F1
lead_frame	0.952	0.843
agency_style	0.952	0.844
density_band	0.952	0.958
rhetorical_mode	0.904	0.816

Interpretation

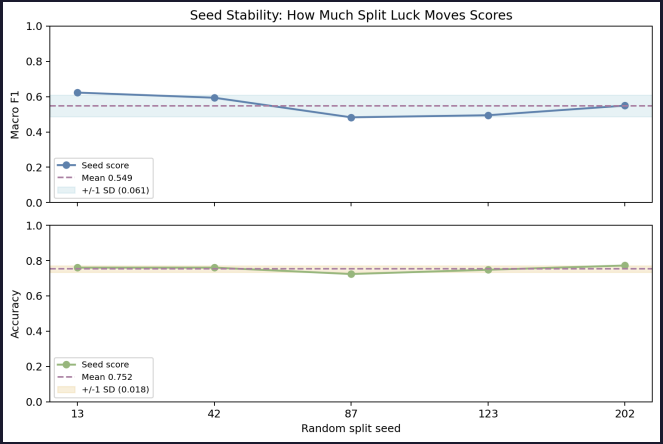
Density and framing are highly recoverable; rhetorical mode remains harder because it depends on broader discourse cues and boundary judgments.

Performance Heatmap



Compact view of performance across structure and style outputs.

Seeding and Split Luck



Why this analysis?

- One split can be lucky/unlucky
- Need expected performance, not one draw

5-seed summary

- Test accuracy: 0.752 ± 0.018
- Test macro F1: 0.549 ± 0.061
- Dev macro F1: 0.571 ± 0.052

Seeding and Split Luck (interpretation)

- Per-seed structure test macro F1 spans **0.483 to 0.623** (spread 0.140).
- Per-seed structure test accuracy spans **0.723 to 0.771** (spread 0.048).
- Macro F1 varies more than accuracy, so class-balance behavior is the more split-sensitive quantity.

Implication

Reporting only one split can overstate confidence. We therefore treat the seed mean as the expected score and the sample standard deviation as an uncertainty descriptor.

How We Quantify Split Variability

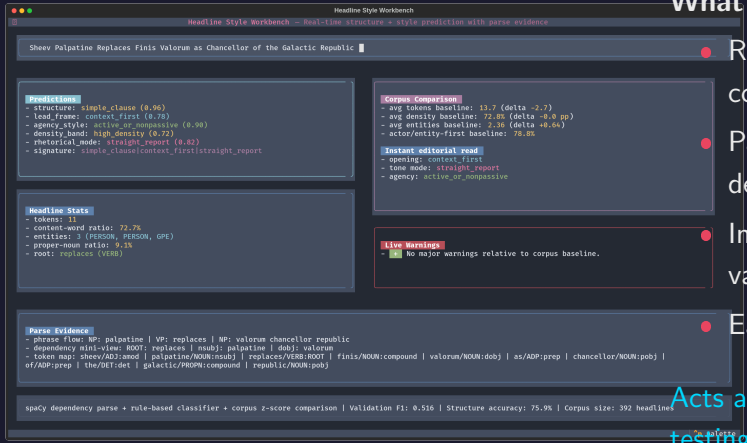
For metric values $m_1, \ldots, m_K$ over K random seeds:

$$\bar{m} = \frac{1}{K} \sum_{i=1}^{K} m_i$$

$$s = \sqrt{\frac{1}{K-1} \sum_{i=1}^{K} (m_i - \bar{m})^2}$$

- $\bar{m}$ estimates expected performance under split randomness
- s measures split sensitivity (evaluation volatility)
- Per-seed trajectories reveal whether variance is smooth or driven by specific difficult splits

Interactive TUI Sandbox for Journalists



What it enables

- Real-time prediction with confidence
- Parse-level evidence for every decision
- Instant iteration over headline variants
- Early warning for risky phrasing

Acts as a sandbox for editorial A/B testing before publication.

Discussion: What Worked and What Remains Hard

Strong signals

- Dominant templates are recoverable
- Multi-dimensional style profiling is informative
- Interpretability is operational, not cosmetic

Difficult regions

- Sparse "other" class behavior
- Coordination vs. simple-clause boundary cases
- Passive edge forms under punctuation/noise

Interpretation: gains are large versus baseline, but minority-structure distinctions remain the principal error source.

Limitations

- Rule-based systems are highly interpretable but less flexible than fully learned models
- Class imbalance can mask minority-class weaknesses
- Some labels remain intrinsically subjective at boundary cases
- Source and language scope limit broad external generalization
- No external lexical expansion in this phase (intentional scope constraint)

Despite these limits, structural signal remains strong and measurable.

Conclusion

Headlines are structured linguistic products

not free-form snippets

Structure test: 0.759 accuracy, 0.594 macro F1

with explicit split-luck quantification

Bottom Line

Interpretable NLP can capture headline form, evaluate it against human gold standards, and provide practical editorial tooling through a real-time sandbox interface.

“Headline form is measurable, predictable, and actionable.”